MUHAMMAD TALHA

AI Engineer | Python Backend Developer

Karachi, Pakistan • m.talha.kamran.pk@gmail.com • github.com/Talha-dev27

PROFESSIONAL PROFILE

AI Engineer and Python developer who designs, evaluates, and deploys production-ready RAG systems, multi-agent frameworks, and intelligent backend architectures. Experienced in building context-aware applications using LangChain and LlamaIndex, optimizing vector embeddings, and creating robust FastAPI microservices for AI deployments. Focused on transforming unstructured raw data into production-grade, deterministic AI workflows.

TECHNICAL CAPABILITIES

AI & REASONING FRAMEWORKS

- LangChain
- LlamaIndex
- OpenAI API
- AI Agents & Tool Use
- Advanced Prompt Engineering

VECTOR DBs & INTERFACES

- Retrieval-Augmented Generation (RAG)
- Pinecone / Chroma / Milvus
- Vector Embeddings & Hybrid Search
- Semantic Data Chunking

BACKEND ENGINEERING

- Python
- FastAPI
- Flask / Django
- REST API Design

DEVELOPER TOOLS & DATA

- Git / GitHub
- Docker & AWS Familiarity
- PostgreSQL & Supabase
- Web Scraping & Data Pipelines

PROFESSIONAL EXPERIENCE

Independent AI Engineer

Self-employed • Remote

- Architect and deploy scalable RAG pipelines using LlamaIndex and LangChain to implement semantic search, document intelligence, and context-rich data synthesis.
- Manage vector store integrations (Pinecone/Chroma) to design low-latency, high-accuracy indices optimized for custom chunking strategies and hybrid retrieval.
- Build high-performance Python backend microservices using FastAPI to securely expose AI model endpoints and coordinate multi-agent reasoning chains.
- Develop custom tools and API connectors to allow LLM agents to autonomously execute web scraping, structured data extraction, and deep competitive analysis.

SELECTED PROJECTS

Lead Radar AI — RAG-Powered Research Engine

Python • LangChain • Pinecone • FastAPI

- Engineered an autonomous intelligence pipeline that ingests web text data, chunks and indexes it into a Pinecone vector database, and applies hybrid search for deep context retrieval.
- Utilized LangChain multi-agent architectures to systematically parse raw data, extract signal insights, and stream structured data objects into production targets.

Studify — Full-Stack Web Application

Vercel • Supabase • Cloud Architecture

- Engineered a cloud-native web application utilizing Vercel for serverless deployments and Supabase for secure, real-time relational database management and user authentication.

EDUCATION & PROFESSIONAL DEVELOPMENT

Cambridge O Level • Focus areas: Computer Science, Mathematics, Physics

Certifications: Advanced Python Programming • AI Systems Mastery • RAG & Vector Architectures